CSE225 Data Structures, 2020(FALL)

HOMEWORK #2 (Deadline: 25.11.2020, 24.00)

Q1. (20 + 40 + 20 + 20 = 100pts)

Consider the C function below!

```
double fact(long i)
{
if (i==1 || i==0) return i;
else return i*fact(i-1);
}
```

a) Assuming that *n* characterizes the input size, **in terms of *n***, what is the number ***S*** of recursive calls? Also give the asymptotic upper bound (*O()*).

b) How would you modify (i.e., rewrite) the function fact so as to eliminate the redundant recursive calls and reduce the number ***S*** of recursive calls?

c) Solve (a) after the modification of the function fact.

d) In terms of *n*, what is the value of sum in the function funcQ2 below? What is its asymptotic upper bound (*O()*)?
In funcQ2 use the first version of fact (i.e., given in (a)).

```
double fact(long i)
{
if (i==1 || i==0) return i;
else return i*fact(i-1);
}

funcQ2()
{
      for (i=1; i<=n; i++)
      sum=sum+log(fact(i));
}
```

Handwriting is mandatory.

Submit your homework into Google Classroom (class code: 7ernnnq).